

1 What is Regularization in Machine learning ? Why is it needed

→ Regularization is a technique used to reduce overfitting by adding a penalty term to the model's loss function

It discourages the model from learning overly complex patterns and helps it generalise better to unseen data

Regularization is needed when a model performs very well on training data but poorly on testing data

2 Difference between Ridge (L2) and Lasso (L1)

(L2)

(L1)

Adds squared coefficient Penalty

Add Absolute coefficient

Reduces coefficients

Can shrink some coefficient exactly to zero

Keeps all feature

Useful when most features are important

Performs better selection automation

Useful when only a few features are important

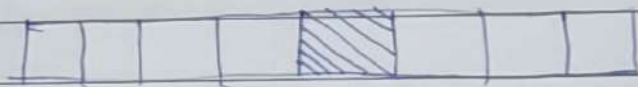
3 What is cross-validation and why is it important?

→ Cross-validation is a technique used to evaluate a model's performance by splitting the dataset into training and testing data. Cross-validation helps detect overfitting, improves model selection and ensures that performance metrics are not dependent on a single train-test split.

4 Explain the following cross-validation techniques

→ K-Fold cross validation

The dataset is divided into k equal parts



Stratified K-Fold cross validation

Similar to K-Fold - but maintaining the same class distribution in every fold

LOOCV

Each sample is used once as a test set while the remaining sample form the training set

Time Series Split:

used for time-dependent data.
Training used past observation,
and preserving chronological order

5 Why are tree-based Models less sensitive to Feature scaling

→ Tree-based models, such as Decision Tree & Random forest, split data based on feature thresholds rather than distance calculation